Supplementary Information

When Better QAOA Schedules Depend on the Simulator: Exact and Cross-Backend Rank Reversals on QOBLIB

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1 Frozen protocol and evaluated schedules

The split and primary promotion gate were written before any candidate was evaluated on `es60fst02`. Training used the complete `es60fst01` and `es60fst03` graphs, validation used `es60fst04`, and `es60fst02` was held out for one confirmatory pass. The gate required a positive paired BKS-rate difference from published LR, a paired bootstrap 95% interval excluding zero, and no material loss of feasible mass. The nonlinear schedule family was

$$\beta_k = \Delta_\beta \left(\frac{p-k+1}{p}\right)^{a_\beta}, \qquad \gamma_k = \Delta_\gamma \left(\frac{k}{p}\right)^{a_\gamma}, \qquad p = 15.$$

Table 1: Schedules frozen before the blind pass. The genome columns are $(\Delta_\beta, \Delta_\gamma, a_\beta, a_\gamma)$.

Method	Δ_β	Δ_γ	a_β	a_γ
Published LR	0.70000	0.40000	1.00000	1.00000
Evolutionary	0.51750	0.77197	1.07734	1.75435
Matched random	0.64247	0.75939	1.77679	0.99172

Three evolutionary replicates used a fixed population of 20 for six generations. Uniform random search received exactly 120 independent candidates per replicate. Every candidate received 256 exact-statevector shots on each training kernel. One champion per search method was selected on validation; test metrics never influenced selection. Samples were only unfolded and checked for independence on the original graph. No repair, greedy fill, local search, or archived solution was used.

The MIS QUBO penalty was $\lambda = 1.5$, with the released maximum quadratic-coefficient normalization. The 186-vertex, 280-edge blind graph was reduced to a 55-qubit, 91-edge kernel. Its native depth-15 circuit contains 1,365 RZZ, 825 RZ, 825 RX, and 55 Hadamard operations before measurement.

2 Blind paired analysis

The confirmatory experiment used 15 paired simulator seeds and 1,000 shots per method and seed. Table 2 reports pooled rates; Table 3 reports paired inference. The confidence interval resamples the 15 paired jobs with 20,000 bootstrap draws. The two-sided sign-flip test enumerates all 2^{15} paired sign assignments. This preserves job-level pairing rather than treating 15,000 individual shots as independent simulator replicates.

Table 2: Blind `es60fst02` results at bond 64 and $\tau_{\mathrm{MPS}} = 10^{-3}$. Each method has 15,000 shots.

Method	BKS hits	BKS rate	BKS−1	Feasible	Quality	N_{95}
Published LR	41	.00273	.11213	.50127	.48962	1,095
Evolutionary	31	.00207	.06813	.89987	.87202	1,449
Matched random	**101**	**.00673**	**.15700**	**.90180**	**.87865**	**444**

Here $N_{95} = \lceil \log(0.05)/\log(1 - p_{\mathrm{BKS}}) \rceil$ is the estimated number of shots required for a 95% chance of at least one BKS sample.

Table 3: Paired method-minus-LR comparisons across 15 jobs. W/T/L denotes wins, ties, and losses.

Method	Metric	Mean Δ	Bootstrap 95% CI	p_{flip}	W/T/L
Evolutionary	BKS	$-.00067$	$[-.00167, .00033]$	.27002	5/2/8
Evolutionary	Feasible	$+.39860$	$[.39133, .40660]$	6.10×10^{-5}	15/0/0
Matched random	BKS	$+.00400$	$[.00247, .00553]$	.000244	13/2/0
Matched random	Near-BKS	$+.04487$	$[.03780, .05140]$	6.10×10^{-5}	15/0/0
Matched random	Feasible	$+.40053$	$[.39266, .40987]$	6.10×10^{-5}	15/0/0
Matched random	Quality	$+.38904$	$[.38140, .39807]$	6.10×10^{-5}	15/0/0

3 Complete 55-qubit sensitivity audit

All sensitivity points are single jobs with 10,000 shots per method, except the primary setting, which pools the 15 confirmatory jobs. Wilson intervals are stored with the artifact. Table 4 exposes the integer hit counts so every displayed rate can be reconstructed.

Table 4: All completed native-circuit MPS settings on the held-out 55-qubit kernel. RS is matched random search.

Bond	τ_{MPS}	Method	Shots	BKS	BKS−1	Feasible	Quality
64	10^{-3}	LR	15,000	41 (.0027)	1,682 (.1121)	7,519 (.5013)	.4896
64	10^{-3}	RS	15,000	101 (.0067)	2,355 (.1570)	13,527 (.9018)	.8787
64	3×10^{-4}	LR	10,000	116 (.0116)	1,542 (.1542)	4,805 (.4805)	.4702
64	3×10^{-4}	RS	10,000	161 (.0161)	1,949 (.1949)	8,129 (.8129)	.7934
64	10^{-4}	LR	10,000	153 (.0153)	1,546 (.1546)	4,809 (.4809)	.4706
64	10^{-4}	RS	10,000	109 (.0109)	1,621 (.1621)	7,351 (.7351)	.7170
96	10^{-3}	LR	10,000	19 (.0019)	1,188 (.1188)	5,055 (.5055)	.4938
96	10^{-3}	RS	10,000	70 (.0070)	1,661 (.1661)	8,925 (.8925)	.8701
96	10^{-4}	LR	10,000	152 (.0152)	1,710 (.1710)	4,799 (.4799)	.4700
96	10^{-4}	RS	10,000	134 (.0134)	1,808 (.1808)	7,121 (.7121)	.6953

The BKS ranking favors RS at 10^{-3} and 3×10^{-4}, but LR at 10^{-4} for both bond caps. Near-BKS, feasible, and quality mass favor RS at all completed points. Raising the bond cap from 64 to 96 does not remove the rank reversal. The attempted bond-128, 10^{-6}, three-method batch exceeded 20 minutes and was terminated without producing a result; it is not treated as a completed point.

4 Exact-state MPS calibration

Complete output states were computed for the real 15-qubit `es60fst01` and 12-qubit `es60fst03` kernels. Probabilities were compared before measurement, eliminating shot noise. For exact distribution p and MPS distribution q, total variation is $\frac{1}{2}\sum_x |p(x)-q(x)|$; fidelity is the squared state overlap. Table 5 gives all bond-128 threshold points.

At each of 10^{-3}, 3×10^{-4}, and 10^{-4}, the state and metric errors are numerically identical at bonds 32, 64, and 128. Bond 16 is also identical except for nonlinear `es60fst01` at 10^{-4}: TV is 0.032981 rather than 0.032493 and BKS error is 0.013332 rather than 0.012953. Thus the threshold dominates once the cap reaches 32 on these donor kernels. The signed LR BKS error crosses zero twice even while fidelity improves, so a monotone global state metric does not certify monotone convergence of a rare observable.

Table 5: Exact-state calibration. p^*_{BKS} is exact BKS probability and $\Delta p_{\mathrm{BKS}} = p^{\mathrm{MPS}}_{\mathrm{BKS}} - p^*_{\mathrm{BKS}}$. NL is the matched-random nonlinear schedule.

Case	Method	τ_{MPS}	Fidelity	TV	p^*_{BKS}	Δp_{BKS}
01	LR	10^{-3}	.980259	.051752	.686100	+.015769
01	LR	3×10^{-4}	.991631	.036703	.686100	+.009153
01	LR	10^{-4}	.995614	.025605	.686100	-.001314
01	LR	10^{-5}	.998979	.008277	.686100	-.000559
01	LR	10^{-6}	.999866	.002659	.686100	+.000153
01	NL	10^{-3}	.952413	.074245	.928984	+.034173
01	NL	3×10^{-4}	.974923	.052601	.928984	+.022460
01	NL	10^{-4}	.986271	.032493	.928984	+.012953
01	NL	10^{-5}	.998091	.008755	.928984	+.002279
01	NL	10^{-6}	.999836	.001448	.928984	+.000220
03	LR	10^{-3}	.987431	.050922	.735826	+.007444
03	LR	3×10^{-4}	.994573	.021938	.735826	+.003234
03	LR	10^{-4}	.997701	.016650	.735826	-.002770
03	LR	10^{-5}	.999535	.004628	.735826	-.000237
03	LR	10^{-6}	.999953	.001805	.735826	+.000204
03	NL	10^{-3}	.976398	.070154	.955300	+.018487
03	NL	3×10^{-4}	.986230	.050596	.955300	+.010784
03	NL	10^{-4}	.994029	.024675	.955300	+.005526
03	NL	10^{-5}	.999340	.005522	.955300	+.000701
03	NL	10^{-6}	.999969	.001053	.955300	+.000133

5 Exact 24-qubit cross-backend ladder

The external extttaves-sparrow-social graph reduces to 24 qubits at the frozen cap 20. Dense statevectors give BKS probabilities 0.32327673 for LR, 0.31113788 for matched random, and 0.42525564 for the evolutionary schedule. Both qubit orderings give the same exact probabilities, as required by their unitary equivalence. The primary exact matched-random-minus-LR effect is -0.01213885.

The independent protocol fixed five MPS points, three schedules, and two orderings before cuTensorNet target execution. All 30 jobs completed without shots or errors. Table 6 reports the full primary analysis. Fidelity and BKS error columns summarize the three cuTensorNet schedules in a cohort; "sign" compares its matched effect with exact.

Table 6: Exact-calibrated 24-qubit cross-backend ladder. Effects are matched-random minus LR BKS probability.

Setting	Order	Aer effect	cuTN effect	Sign	Min fidelity	Max TV	Max BKS err.
$64/10^{-3}$	Sorted	+.138479	-.008287	Yes	.798088	.240005	.013000
$64/10^{-3}$	Spectral	+.105273	+.016849	No	.698032	.282899	.115859
$128/10^{-4}$	Sorted	+.001345	-.007811	Yes	.942677	.099084	.004925
$128/10^{-4}$	Spectral	+.001817	+.004293	No	.812569	.181599	.047608
$128/10^{-12}$	Sorted	-.003717	-.009955	Yes	.944334	.089741	.004537
$128/10^{-12}$	Spectral	-.011907	+.016699	No	.807278	.197239	.060079
$1024/10^{-4}$	Sorted	+.000856	-.012031	Yes	.997276	.010526	.000128
$1024/10^{-4}$	Spectral	+.001817	-.012207	Yes	.997210	.010731	.000150
$1024/10^{-5}$	Sorted	-.012235	-.012156	Yes	.999997	.000453	.000019
$1024/10^{-5}$	Spectral	-.012831	-.012324	Yes	.999982	.001240	.000229

The comparison separates three phenomena. First, cutoff dominates cap within each implementation but not identically. Second, exact-equivalent orderings can cross the zero-effect boundary after truncation. Third, a nominal MPS setting is not a cross-backend accuracy contract: at $1024/10^{-4}$ cuTensorNet is sign-correct in both orderings while Aer is sign-wrong in both. The selected $1024/10^{-5}$ point is the only one sign-correct across exact, both backends, and both orderings.

A separate post-hoc certificate audit applies the event-probability inequality $|\widetilde{\Delta} - \Delta| \le \mathrm{TV}_{\mathrm{LR}} + \mathrm{TV}_{\mathrm{matched}}$. Of 20 backend/setting/ordering cohorts, 11 have the realized exact sign, five are certified by the TVD margin, and only two are certified by substituting the fidelity bound $\mathrm{TV} \le \sqrt{1-F}$. Every certified cohort is sign-correct. The certificate is therefore conservative by design and is not presented as a replacement for the pre-registered exact comparison.

6 Frozen five-case exact replication

The cross-case protocol was frozen on 11 August 2026 before any new target row was generated. It retains the five MPS settings from the 24-qubit ladder and adds four QOBLIB instances with already completed exact adjudication. All cases use depth 15, the same three frozen schedules, sorted and spectral orderings, and no repair. Aer MPS and cuTensorNet each contribute 150 rows. Of these, 120 per backend are newly executed; the 30 per backend from `aves-sparrow-social` are reused only after validating their recorded SHA-256 hashes. The final analysis therefore contains 300 unique rows and 100 matched-random-versus-LR backend/setting/ordering cohorts.

For event A, write $\Delta = p_i(A) - p_j(A)$ and $\widetilde{\Delta} = q_i(A) - q_j(A)$. The triangle inequality and the defining event characterization of total variation give

$$\begin{aligned} |\widetilde{\Delta} - \Delta| &= |[q_i(A) - p_i(A)] - [q_j(A) - p_j(A)]| \\ &\le |q_i(A) - p_i(A)| + |q_j(A) - p_j(A)| \\ &\le \mathrm{TV}(q_i, p_i) + \mathrm{TV}(q_j, p_j). \end{aligned}$$

Thus $|\Delta| > \mathrm{TV}_i + \mathrm{TV}_j$ is a sufficient condition for sign preservation. It is intentionally conservative: failure to certify does not imply an incorrect sign.

Table 7: Five-case exact replication. "Sign" counts correct signs across two backends, five settings, and two orderings. "Cert." counts exact-margin TVD certificates. Backend agreement is across five settings and two orderings.

Case	Qubits	Exact effect	Sign	Cert.	Backend	First universal
karate	3	+.086412	20/20	20/20	10/10	released
chesapeake	7	-.134214	20/20	20/20	10/10	released
football	7	+.019269	20/20	18/20	10/10	confirm
ibm32	18	-.246123	20/20	14/20	10/10	$1024/10^{-4}$
aves-sparrow-social	24	-.012139	11/20	5/20	5/10	none
All	3–24	—	91/100	77/100	45/50	—

All 100 observed effect errors satisfy the bound; the largest numerical excess is 6.6×10^{-16}. The normalized bound is below one in 77 cohorts and all 77 have the correct sign (Clopper–Pearson 95% CI $[0.953, 1]$). Among the 23 uncertified cohorts, 14 signs are correct and nine are wrong (95% CI for the correct fraction $[0.385, 0.803]$). The resulting 2×2 Fisher exact test is $p = 4.30 \times 10^{-7}$. Because settings within a case share circuits and are not independent population draws, this test is reported only as a descriptive association. The deterministic implication below ratio one follows directly from the inequality and does not require statistical independence.

The certificate is less conservative than substituting the fidelity relation $\mathrm{TV} \le \sqrt{1-F}$: 77 cohorts are TVD-certified but only 58 are fidelity-certified. The per-case results also show why qubit count alone is not an error model. The 18-qubit `ibm32` case has a large exact margin and preserves all 20 signs, whereas the 24-qubit `aves-sparrow-social` case has the smallest margin and contains all nine failures. Three uncertified cohorts among the first four cases are correct, confirming that the certificate is sufficient rather than necessary.

7 55-qubit independent cuTensorNet audit

We exported the same native 55-qubit circuits to NVIDIA cuTensorNet 26.6 and decoded samples with the unchanged QOBLIB reduction map. An independent exact contraction on the 12- and 15-qubit donor kernels agreed with Qiskit statevectors to unit fidelity for both schedules and both qubit orderings. On the 55-qubit kernel, the exact sampler failed during contraction planning with an internal backend error. We therefore report the successful cuTensorNet MPS sweep, including the failed attempt in the audit log.

A Fiedler ordering of the interaction graph reduced bandwidth from 47 to 10 and maximum cut width from 42 to 9. It is an exact relabeling: samples were inverse-permuted before decoding. This reduced the bond-32, 100-shot runtime from 428–463 seconds per schedule in sorted order to 65–69 seconds. Table 8 gives the completed independent sweep. Near means objective at least BKS−1.

At 10^{-3}, NL has 15 versus 6 BKS hits (Fisher two-sided $p = .078$); at 10^{-4} the counts are 11 versus 12 ($p = 1.0$). Thus the independent backend reproduces loss of the apparent NL advantage under tighter truncation, but not a statistically resolved reversal. Feasible and near-BKS mass favor NL at both thresholds.

Table 8: Independent cuTensorNet MPS results on `es60fst02`. NL is the frozen matched-random nonlinear schedule.

Order	Bond	τ_{MPS}	Method	Shots	BKS	Near	Feasible	Seconds
Sorted	32	10^{-3}	LR	100	0	.000	.250	428.3
Sorted	32	10^{-3}	NL	100	0	.000	.700	463.3
Fiedler	32	10^{-3}	LR	100	0	.030	.440	64.7
Fiedler	32	10^{-3}	NL	100	0	.040	.740	68.7
Fiedler	64	10^{-3}	LR	200	1 (.0050)	.020	.440	113.6
Fiedler	64	10^{-3}	NL	200	1 (.0050)	.065	.725	117.0
Fiedler	128	10^{-3}	LR	5,000	6 (.0012)	.0424	.4840	228.7
Fiedler	128	10^{-3}	NL	5,000	15 (.0030)	.0820	.6232	252.7
Fiedler	128	10^{-4}	LR	5,000	12 (.0024)	.0764	.4838	222.2
Fiedler	128	10^{-4}	NL	5,000	11 (.0022)	.0782	.6352	260.1

8 Strong classical MIS controls

The complete 186-vertex graph was also solved directly with two classical controls. SciPy/HiGHS mixed-integer programming certified the optimum 88 with zero reported gap in 0.036 seconds. The stochastic control repeatedly picks a minimum residual-degree vertex with random tie breaking. It uses no archived solution, parameter training, or repair after sampling. We ran 15 independent replicates of 1,000 starts on both the full graph and the released 55-variable reduced kernel.

Table 9: Classical controls versus the strongest released-setting QAOA row. QAOA time is the measured batch wall time divided equally among three methods.

Method	Trials	BKS hits	BKS rate	Near	Feasible	Seconds
HiGHS exact MILP	1	1	1.0000	1.0000	1.0000	0.036
Full-graph greedy	15,000	1,133	.07553	.31527	1.0000	19.95
Kernel greedy	15,000	6,063	.40420	.96873	1.0000	2.21
QAOA NL, Aer MPS	15,000	101	.00673	.15700	.9018	183.04

Full-graph and kernel greedy achieve 11.2 and 60.0 times the QAOA BKS rate, respectively, while also running faster. The exact solver is faster still on this instance. These controls rule out a competitive or quantum-advantage claim for the present experiment; the contribution is the cross-backend diagnosis of rare-event instability.

9 Audit trail and artifact map

The first validation and blind MPS pass used Qiskit transpilation at optimization level zero. Inspection of the released QOBLIB workflow showed that its Aer/MPS baseline receives native RZ/RZZ/RX circuits. Because basis decomposition changes where MPS truncation is applied, the first pass was preserved as a protocol audit and excluded from the main result. Champions were not reselected after this discovery; the corrected pass reevaluated the already frozen schedules. Training was exact-statevector simulation, where the gate representation does not change the unitary.

Table 10: Machine-readable artifact map. Paths are relative to the artifact root.

Purpose	Artifact
Frozen decisions	FROZEN_PROTOCOL.md
Deviation log	PROTOCOL_DEVIATIONS.md
Blind raw jobs	results/blind_test.json
Paired inference	results/analysis_summary.json
Sensitivity counts	results/mps_sensitivity.csv
Exact states and metrics	results/exact_mps_calibration.json
Compact exact table	results/exact_mps_calibration.csv
Classical controls	results/classical_baselines.json
Extended comparison	results/extended_comparison.json
cuTensorNet sweep	results/cutensornet_sweep.csv
Independent exact check	results/cutensornet/small_exact_validation.json
Independent attempt log	results/cutensornet/ATTEMPTS.md
24q exact protocol	EXACT_EXTENSION_PROTOCOL.md
24q MPS ladder protocol	MPS_LADDER_PROTOCOL.md
Independent ladder protocol	INDEPENDENT_LADDER_PROTOCOL.md
Cross-case protocol	CROSS_CASE_REPLICATION_PROTOCOL.md
Exact reference manifest	results/mps_ladder/exact_references.json
Aer ladder	results/mps_ladder/mps_ladder.json
cuTensorNet 30-job ladder	results/independent_ladder/mps_jobs.json
Cross-backend analysis	results/independent_ladder/analysis.json
Rank certificates	results/rank_certificates.json
Cross-case backend rows	results/cross_case_replication/*_jobs.json
Cross-case analysis	results/cross_case_replication/analysis.json
Cross-case paper statistics	results/cross_case_replication/paper_statistics.json
Transpiled audit	results/blind_test_transpiled_audit.json
Integrity tests	test_*.py
Checksums and provenance	artifact_manifest.json

The integrity and unit tests check the published ramp formula, split disjointness, the 186-to-55 reduction, disabled repair, balanced 15,000-shot blind artifacts, the 56-row exact-calibration matrix, tight-threshold convergence, axis conversion, literal decoding, and frozen design size. They also check the completeness and validity of the classical controls and both independent backend artifacts, frozen 300-row design, TVD computations, and NumPy-safe JSON serialization. The analysis scripts regenerate the CSV tables, figures, statistical summaries, and reports from the JSON artifacts.

10 Post-freeze breadth and hardware-readiness addendum

The original five-case replication is unchanged. An independently documented 50-instance eligibility screen admitted eight graphs; the three additional eligible cases were `es60fst01`, `es60fst03`, and `mammalia-kangaroo-interactions`. Their subsequent single-backend Aer pilot completed 18 exact rows and 108 seeded MPS rows (54,000 shots), with 24/24 candidate-versus-LR effect signs preserved. These observations are not pooled with the original deterministic two-backend cohorts and do not establish population-level prevalence of reversals.

10.1 Finite-shot planning

The 9 September protocol uses the already observed exact BKS probabilities for sorted ordering and matched-random versus LR on each of the five original cases. It is a post-hoc planning analysis, not an independent validation set. For each of five budgets per schedule (500, 2,000, 10,000, 50,000, 200,000), 10,000 independent pairs of binomial counts are generated with seed 20260909. Each arm receives a two-sided Clopper–Pearson interval with error probability $0.05/(2 \cdot 5) = 0.005$. Subtracting the arm intervals gives an effect interval; a sign is reported only if this excludes zero. A union bound controls the familywise error at 0.05 across the five comparisons at a *fixed* budget. The five budgets are alternative planning scenarios, not repeated peeking. No drift robustness or cross-budget simultaneous guarantee is asserted.

For `football`, correct sign resolution is 53.43% at 10,000 shots per arm and 100% in the 10,000 replicates at 50,000 shots. The 24-qubit small-gap case needs substantially more shots (full rates are in the machine-readable artifact). Zero wrong resolved signs were observed across the 250,000 simulated comparisons. This observation is not a proof of zero error: the statistical guarantee comes from interval coverage and the union bound. Increasing shots cannot remove a simulator's systematic bias.

10.2 Synthetic noise and circuit mapping

Six archived circuits (three cases of three or seven qubits, two schedules) are hash-checked and transpiled to a bidirectional line, basis $\{R_Z, \mathrm{SX}, X, \mathrm{CX}\}$, optimization level one, initial identity layout, and seed 20260909. Final logical-to-physical permutations are inverted before applying the frozen BKS decoder. Both schedules have respectively 156, 1,232, and 708 CX gates on `karate`, `chesapeake`, and `football`. Zero-noise BKS probabilities agree with archived exact references within 10^{-9}.

The channel after each CX is $\mathcal{D}_\lambda(\rho) = (1-\lambda)\rho + \lambda I/4$, with $\lambda \in \{0, 10^{-4}, 10^{-3}, 0.003, 0.01\}$; the corresponding one-qubit channel after SX and X has parameter $\lambda/10$. RZ gates are noise-free in this synthetic model. For nonzero λ, a 1% independent symmetric bit-flip channel is applied to the output distribution. All 30 jobs use exact density-matrix evolution, not sampled trajectories. These parameters are channel parameters, not reported device gate infidelities. The noiseless-to-smallest-noise comparison changes both gate and readout noise; it is not an isolation of either contribution.

Table 11: Matched-random minus LR BKS probability under synthetic noise.

Case	No noise	$\lambda = 10^{-3}$	$\lambda = 0.01$
karate	+0.086412	+0.065200	+0.001615
chesapeake	−0.134214	−0.047042	−0.000319
football	+0.019269	−0.001387	−0.003512

The football sign reversal demonstrates why a hardware ranking cannot by itself adjudicate the ideal ranking. This local model is not a calibration of a named device and is not evidence for a new physical mechanism. Actual hardware validation requires a target-specific transpilation, recorded calibration, fixed measurement budget, interleaved schedules, and explicit accounting for time dependence. No QPU job was submitted for this addendum.

The repository-level `research_package` directory contains the frozen planning protocol, runner, six compiled circuit files, unit tests, and `results/readiness.json` with source and circuit hashes.